Equation Detective Case Files

Unit 7: Equation Detective Agency · Lesson 7-1 · Flagship · EduWonderLab

UNIT 7 · LESSON 7-1 ★ FLAGSHIP · TEACHER NOTES — **Standard 6.EE.6**

FORMAT **Case-file mystery**	TIME **30–35 min**
GROUPING **Teams of 3–4**	TOPIC **Write Equations (Flagship)**
STANDARD **6.EE.6**	PREP LEVEL **Medium — print case-file folders with clue cards**

Learning goal *I can translate clues into equations and write my own equation mystery.*

Teacher setup

- Print one case-file set per team. Each clue describes the same mystery number.
- Teams translate each clue into an equation and check it gives the same answer.
- Add “red herring” details that do not affect the math.

Materials

Case-file folders, Clue cards, Equation template, Pencil.

Differentiation & language support

- Equation template: words → variable → equation (SPED).
- Sentence stem: “The clue says ___, so the equation is ___.”
- ESOL: cross out the red-herring detail before writing the equation.

Key vocabulary (English / Español):

Term	Español	What it means
clue	pista	information that helps solve the mystery
red herring	pista falsa	a detail that does not matter

Quick teacher check: *All clues should solve to 12; check the final-case equation.*

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Name: **Date:** **Class:**

Your goal: *I can translate clues into equations and write my own equation mystery.*

What you need

Case-file folders, Clue cards, Equation template.

How to play

1. Open a clue and ignore any red-herring detail.
2. Write an equation for the clue.
3. Solve to find the mystery number.
4. Check that every clue gives the same number.

Case Clues

Turn each clue into an equation. They all describe the same mystery number.

1. Clue A: A mystery number plus 8 equals 20.	**2.** Clue B: Three times the number is 36.
3. Clue C: The number doubled, then add 5, equals 29.	**4.** Clue D: Half the number is 6.
5. Clue E: The lab was cold (ignore). The number minus 4 is 8.	**6.** Final Case: Write a new equation mystery whose answer is 10.

Detective log (equation + solution for each clue):

Explain your thinking

Explain how a red herring is different from a real clue in solving the case.

Sentence frame: *The clue says ___, so the equation is ___, and the number is ___.*

★ **Challenge:** *Write three different clues that all point to the mystery number 15.*

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. Clue A: A mystery number plus 8 equals 20. **Answer: $x + 8 = 20 \rightarrow x = 12$**	**2.** Clue B: Three times the number is 36. **Answer: $3x = 36 \rightarrow x = 12$**
3. Clue C: The number doubled, then add 5, equals 29. **Answer: $2x + 5 = 29 \rightarrow x = 12$**	**4.** Clue D: Half the number is 6. **Answer: $x \div 2 = 6 \rightarrow x = 12$**
5. Clue E: The lab was cold (ignore). The number minus 4 is 8. **Answer: $x - 4 = 8 \rightarrow x = 12$**	**6.** Final Case: Write a new equation mystery whose answer is 10. **Answer: e.g., $x + 3 = 13 \rightarrow x = 10$**

Teaching notes & look-fors

- Every clue solves to the mystery number 12.
- Red herrings (Clue E's cold lab) do not change the equation.